Résumé: Dr. Matthias Mittner (born Ihrke)

Contact Information

Matthias Mittner (born Ihrke)

Max-Planck Institute for Experimental Medicine
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Personal Information

Date of birth: September 1, 1982
Place of birth: Eberswalde, Germany
Nationality: german
Marital Status: Married to Lilli Mittner
Children: Son Kuno, Daughter Josefine

Education

Dr. phil. (summa cum laude) — 21.12.2011
University of Frankfurt, Germany
Thesis: "Binding and Retrieval of Stimulus-Response Episodes in the Negative Priming Paradigm"

B. Sc. in Computer Science — 2005/07 – 2008/04
Institute of Computer Science, University of Göttingen, Germany
Thesis: "Single Trial Estimation and Timewarped Averaging of Event-Related Potentials"

Diploma in Psychology — 2005/07 – 2007/11
Department of Psychology, University of Göttingen, Germany
Thesis: "Negative Priming and Response-Relation: Behavioural and Electroencephalographic Correlates"

Student at the University of Oslo — 2004/08 – 2005/06
Oslo, Norway
Erasmus-Scholarship; studies of computer science and psychology

B. Sc. in Psychology (equivalent) — 2002/09 – 2004/07
Department of Psychology, University of Göttingen, Germany
Thesis: "An Adaptive Approach to Memory-Span Tasks"

High-School Diploma — 2002/06
Alexander von Humboldt Gymnasium, Eberswalde, Germany
main subjects physics and mathematics

Employments

Post-Doctoral Researcher — 2013/04 – present
Cognitive Science Center Amsterdam, Universiteit van Amsterdam, The Netherlands
Cognitive Neuroscience (Birte Forstmann)

Post-Doctoral Researcher 2012/09 – 2013/03
Max-Planck Institute for Experimental Medicine, Göttingen, Germany
Theoretical Neuroscience (Robert Gütig)

PhD-Student 2008/10 – 2012/01
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)
Doctoral Program: Theoretical and Computational Neuroscience (PTCN)

Guest Researcher 2008/03 – 2008/09
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
cognitive modelling and data-mining (EEG)

Guest Researcher 2007/11 – 2008/02
Max-Planck Institute for Informatics, Saarbrücken, Germany
section for computer graphics; optical modelling and psychophysics

Research Assistant 2005/09 – 2007/10
Bernstein Center for Computational Neuroscience, Göttingen, Germany
cognitive modelling; psychophysical experiments

Student Assistant 2003/09 – 2004/08
Departement for Developmental Psychology, University of Göttingen, Germany
computer administration and development of experimental software

Scientific Activities

Professional Memberships
Program Committee Member "International Conference on Intelligent Data Engineering and Automated Learning" 2010 and 2011
Member of the Research Priority Program Brain and Cognition at the Cognitive Science Center Amsterdam (CSCA)

Organizing
Organizer for Methods-course "Introduction to Python programming" (2012)
Organizer for Methods-course "Matlab and Python programming" (2011)

Conferences
Talks and Posters at national and international conferences (CNS 06/07, BCCN symposia, TeaP, ICP2008, IDEAL, IJMC, NWG) – please see my website for a complete list

Advanced Courses
Course on Computational Neuroscience at the Max-Planck-Institute for Dynamics and Self-Organization 2005

DGPs Course on Recording and Analysis of Eye-Movements (Niels Galley), Cologne 2008

Reviewing
Reviewer for Spring Conference on Computer Graphics 2008 (SCCG08)
Ad-hoc Reviewer for "Cognitive Neuropsycholoy"
IDEAL 2010/2011 (PC-member)

Publications

Journal Articles

Mittner, M. (2013). Functional Integration of Large-Scale Brain Networks. *The Journal of Neuroscience*, 33(48):18710-18711

Schrobsdorff H., *Ihrke M.*, Behrendt J., Hasselhorn M., Herrmann J.M. (2012). Inhibition in the Dynamics of Selective Attention: An Integrative Model for Negative Priming. *Frontiers in Psychology* doi:10.3389/fpsyg.2012.00491

Ihrke M., Behrendt J., Schrobsdorff H., Visser I., Hasselhorn M. (2012, accepted). Negative Priming Persists in the Absence of Response-Retrieval. *Experimental Psychology.*

Schrobsdorff H., *Ihrke M.*, Behrendt J., Herrmann J. M., Hasselhorn M. (2012). Identity Negative Priming: A Phenomenon of Perception, Recognition or Selection? *PLoS ONE 7* (3), `doi:10.1371/journal.pone.0032946`.

Ihrke, M. & Behrendt, J. (2011). Automatic generation of randomized trial sequences for priming experiments. *Frontiers in Psychology, 2* (255), `doi:10.3389/fpsyg.2011.00225`.

Ihrke, M. & Brennen, T. (2011). Sharing one biographical detail elicits priming between famous names: Empirical and computational approaches. *Frontiers in Psychology*, 2 (75), `doi:10.3389/fpsyg.2011.00075`.

Ihrke, M., Schrobsdorff, H. & Herrmann, J.M. (2011). Recurrence-Based Estimation of Time-Distortion Functions for ERP Waveform Reconstruction. *International Journal of Neural Systems*, 21 (1), pp. 65-78.

Ihrke, M., Behrendt, J., Schrobsdorff, H., Herrmann, J.M. & Hasselhorn, M. (2011). Response Retrieval and Negative Priming: Encoding and Retrieval Specific Effects. *Experimental Psychology*, 58 (2), pp. 154–161.

Behrendt, J., Gibbons, H., Schrobsdorff, H., *Ihrke, M.*, Herrmann, J.M. & Hasselhorn, M. (2010). Event-Related Brain Potential Correlates of Idendity Negative Priming From Overlapping Pictures. *Psychophysiology*, 47 (5), pp. 921–930.

Ritschel, T., Ihrke, M., Frisvad, J.R., Coppens, J., Myszkowski, K. & Seidel, H.P. (2009). Temporal Glare: Real-Time Dynamic Simulation of the Scattering in the Human Eye. *Computer Graphics Forum.* 28 (2). pp. 183-192.

Ritschel, T., Smith, K., Ihrke, M., Grosch, T., Myszkowski, K., & Seidel, H. P. (2008). 3D Unsharp Masking for Scene Coherent Enhancement. *ACM Transactions on Graphics (Proceedings SIGGRAPH 2008).* 27 (3).

Schrobsdorff, H., Ihrke, M., Kabisch, B., Behrendt, J., Hasselhorn, M., Herrmann, J. M. (2007). A Computational Approach to Negative Priming. *Connection Science*, *19* (3), 203-221.

Book Chapters

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2009). Denoising and Averaging Techniques for Electrophysiological Data. In Wennberg, R. & Perez-Velazquez, J. L. (Eds.), *Coordinated Activity in the Brain: measurements and relevance to brain function and behaviour.* Springer.

Peer-Reviewed Conference Proceedings

Schrobsdorff H., *Ihrke M.*, Herrmann J.M. (2012). Modeling Structure and Dynamics of Selective Attention. *Advances in Intelligent Systems and Computing* (Biologically Inspired Cognitive Architectures 2012), 196, pp. 287–295.

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2009). Recurrence-Based Synchronization of Single Trials for EEG-Data Analysis. *Lecture Notes in Computer Science* 5788, Intelligent Data Engineering and Automated Learning - IDEAL 2009. pp. 118-125.

Ihrke M., Ritschel, T. , Smith, K., Grosch, T., Myszkowski, K., & Seidel, H. P. (2009). A Perceptual Evaluation of 3D Unsharp Masking. *Proceedings Human Vision and Electronic Imaging XIII.*

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2008). Compensation for Speed-of-Processing Effects in EEG-Data Analysis. *Lecture Notes in Computer Science (Proceedings IDEAL 2008).* 5326, pp. 354-361.

Yoshida, A., Ihrke, M., Mantiuk, R. & Seidel, H. P. (2008). Brightness of the Glare-Illusion. *Proceedings of ACM Symposium on Applied Perception in Graphics and Visualization.* pp. 83-90.

Teaching

Introduction to Python Programming (A123) — winter term, 2011/2012
GGNB methods course (introductory level), University of Göttingen

Gestaltung von Lehr-Lernprozessen — summer term, 2010
Introductory seminar to the psychology of learning and teaching for students pursuing teaching certification; University of Göttingen

Matlab and Python Programming (A109) — winter term, 2010/2011
GGNB methods course (introductory level); University of Göttingen

Students (Supervision)

Ira Beckmann — 2010/03
Diploma-Thesis: "Negatives Priming als Identifikations- oder Selektionsphänomen"; University of Göttingen

Scholarships

GGNB Fellowship — 2009/05–2012/01
Full–Time PhD Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

GGNB Bridging Fund Scholarship — 2008/10–2008/12
Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

DFG Scholarship — 2009/01–2009/04
Full–Time Scholarship "Passungsverhältnisse schulischen Lernens: Verstehen und Optimieren"

Erasmus Scholarship — 2004/07–2005/08
Scholarship for studies abroad, Oslo (UiO), Norway

Skills

Programming Languages
Very Good: C, C++, Python, Matlab/Octave, R
Good: Perl, Java, SQL, PHP, Presentation
Moderate: F77, Assembler, Objective C, Bash, Javascript

Computing
Parallel Programming (MPI), GPU Programming (OpenGL), Unix/Linux Administration, LaTeX, HTML/CSS

Languages	*German* (native) *English* (fluently written, read and spoken) *Norwegian* (fluently read and spoken, basic writing skills) *French* (basic skills) *Dutch* (basic skills)
Sports	judo, rock-climbing, hiking, table-tennis, football